ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Cave - arch - stack	Erosion sequence
2	**(c)** Dense population and weak buildings	Population density
3	**(b)** mitigation measure	Preparedness
4	**(a)** Both A and R are true and R is the correct explanation of A.	Convergent boundary
5	**(c)** GLOF	Moraine dam
6	**(a)** wave erosion	Stacks and cliffs
7	**(d)** Sand dune	Wind deposition
8	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R describes one type, not full reason
9	**(a)** Both A and R are true and R is the correct explanation of A.	Collision and compression of the crust at a convergent boundary folded the sediments into the Himalaya.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but unrelated: one deals with plates moving apart, the other with plates moving together.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Used contouring and bunding (1); Built dams and canals (1).	1+1
12	Slopes limit farming (1); Steps reduce erosion and hold water (1).	1+1
13	Cuts and weakens slopes (1); Disturbs drainage (1).	1+1
14	Wind direction varies (1); Sand supply and vegetation differ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Waves constantly erode and deposit (1); Human activity removes sand (1); Sea-level rise (1).	1+1+1
16	Desert: physical due to temperature swings (1); Forest: chemical and biological due to moisture (1); Climate decides (1).	1+1+1
17	Plant windbreaks (1); Control overgrazing (1); Improve farming (1).	1+1+1
	OR: Hot material rises (1); Cool material sinks (1); Plates dragged along (1).	1+1+1
18	Store water as ice (1); Release slowly in summer (1); Feed rivers (1).	1+1+1
	OR: Upper course: erosion (1); Middle: erosion and deposition (1); Lower: deposition (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The Ring of Fire (1). (ii) Convection currents in the mantle (1). (iii) Convergent boundary (1). (iv) The Indian plate collided with the Eurasian (continental) plate at a convergent boundary; neither could sink easily, so the crust crumpled and folded upward, forming fold mountains (2).	1+1+1+2
20	Contouring, bunding, terracing (1); Zabo system (1); Check dams (1); Modern monitoring and policy (1); Combined approach works best (1).	1+1+1+1+1
	OR: Rain adds weight (1); Slope increases risk (1); Deforestation and mining (1); Poor drainage (1); Preparedness and land-use planning (1).	1+1+1+1+1

Mistakes to avoid:

- **Weathering** breaks rocks in place; **erosion** carries material away by water, wind or ice. They are not the same process.
- Convergent boundary = plates move towards each other (Himalaya); divergent = move apart (mid-ocean ridges); transform = slide past (San Andreas Fault).

- Oceanic crust is thinner (about 5-7 km) than continental crust (about 30-40 km); the mantle is the thickest layer and the inner core the densest.
- Sea-coast sequence is cave, arch, stack; do not mix glacial features (cirque, fjord, U-valley, moraine, arete) with river features (V-valley, meander, oxbow lake, delta).
- A delta is a deposit at the river mouth in the lower course; an oxbow lake is a cut-off meander; waterfalls and V-shaped valleys belong to the upper course.
- Barchans and ventifacts are desert (wind) features and stalactites are limestone features; the Zabo system is from Nagaland and terraces or bunds help check erosion.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Divergent
2	**(c)** fjord	Glacial valley flooded
3	**(d)** A is false but R is true.	No crust destroyed; A false
4	**(c)** Overgrazing and drought	Loose soil
5	**(c)** cutting the slope and poor drainage	Human
6	**(c)** Sea breeze	Winds
7	**(a)** Both A and R are true and R is the correct explanation of A.	Outburst
8	**(a)** Lateral moraines of the two join	Merge
9	**(c)** A is true but R is false.	Terraces are level steps that slow the run-off, so R is false.
10	**(d)** A is false but R is true.	At a transform boundary (e.g. San Andreas Fault) crust is neither created nor destroyed, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Roots and cover are lost (1); Soil is loose and rain runs off fast (1).	1+1
12	Silk Route through deserts (1); Passes like Khyber linked regions (1).	1+1
13	Landslides, avalanches and GLOFs (1); Preparedness saves lives (1).	1+1
14	Melt feeds rivers (1); Glacial lakes and floods affect flow (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Stalactites and stalagmites attract visitors (1); Overuse damages formations (1); Needs protection (1).	1+1+1
16	Barchan: one wind direction (1); Star: winds from many directions (1); Sand supply matters (1).	1+1+1
17	Sudden collapse of ground (1); Damage to buildings and roads (1); Groundwater pumping increases risk (1).	1+1+1
	OR: Slows run-off (1); Water infiltrates (1); Recharges groundwater (1).	1+1+1
18	Silt makes soil fertile (1); Flood destroys crops and homes (1); Needs management (1).	1+1+1
	OR: Coastal: cliffs, stacks (1); Glacial: cirques, aretes (1); Agents are waves and ice (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Upper course (1). (ii) Oxbow lake (1). (iii) Delta (1). (iv) In the plains the river erodes the outer bank where the current is fast and deposits silt on the inner bank where it is slow; the curve grows into a winding bend or meander (2).	1+1+1+2
20	Cliffs and platforms (1); Arches and stacks (1); Beaches and tourism (1); Loss of land (1); Sea walls and planting (1).	1+1+1+1+1
	OR: Glaciers melt faster (1); GLOF risk rises (1); Heavy rain triggers landslides (1); Dust storms in dry areas (1); Preparedness needed (1).	1+1+1+1+1

Mistakes to avoid:

- **Weathering** breaks rocks in place; **erosion** carries material away by water, wind or ice. They are not the same process.
- Convergent boundary = plates move towards each other (Himalaya); divergent = move apart (mid-ocean ridges); transform = slide past (San Andreas Fault).
- Oceanic crust is thinner (about 5-7 km) than continental crust (about 30-40 km); the mantle is the thickest layer and the inner core the densest.
- Sea-coast sequence is cave, arch, stack; do not mix glacial features (cirque, fjord, U-valley, moraine, arete) with river features (V-valley, meander, oxbow lake, delta).

- A delta is a deposit at the river mouth in the lower course; an oxbow lake is a cut-off meander; waterfalls and V-shaped valleys belong to the upper course.
- Barchans and ventifacts are desert (wind) features and stalactites are limestone features; the Zabo system is from Nagaland and terraces or bunds help check erosion.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Slow run-off
2	**(d)** A is false but R is true.	A false, R true
3	**(a)** Lower course	Deposition
4	**(d)** Earthquake - weathering	Internal vs external
5	**(d)** sandblasting by wind	Wind abrasion
6	**(a)** Both A and R are true and R is the correct explanation of A.	Deposition
7	**(a)** Barchan	Wind dune
8	**(a)** Both A and R are true and R is the correct explanation of A.	Karst
9	**(a)** Both A and R are true and R is the correct explanation of A.	Deep, sheltered water lets large ships anchor safely.
10	**(a)** Both A and R are true and R is the correct explanation of A.	The dissolving action makes caves, with stalactites hanging from the roof.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	India sits on this plate pressing against Eurasian plate (1); Stress along the Himalaya causes quakes (1).	1+1
12	Glaciers carved valleys there (1); Sea floods them after ice melts (1).	1+1
13	River loses energy (1); Silt is dropped on floodplains and delta (1).	1+1
14	Meander loop is cut off (1); Water remains in the abandoned loop (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sudden event (1); Allows evacuation (1); Saves lives and livestock (1).	1+1+1
16	Subduction melts crust to magma (1); Transform boundary only slides (1); No magma produced (1).	1+1+1
17	Himalaya protective barrier (1); Khyber Pass for exchange (1); Rivers for farming (1).	1+1+1
	OR: Monitor the lake and install sensors (1); Controlled drainage and warning system (1); Community drills (1).	1+1+1
18	Steep slopes - landslides (1); Snow - avalanche (1); Glacial lakes - GLOF (1).	1+1+1
	OR: Soil saturated and heavy (1); Roots absent to hold soil (1); Landslide risk (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Physical weathering (1). (ii) Biological weathering (1). (iii) Terraces (1). (iv) Both slow down the run-off of rainwater across the slope, so water has time to soak in and less soil is carried away (2).	1+1+1+2
20	Rivers: deltas, waterfalls for farming and power (1); Glaciers: U-valleys, fjords for tourism and harbours (1); Wind: dunes as barriers and sand source (1); Each has hazards (1); Plan land use wisely (1).	1+1+1+1+1
	OR: Bhukampa terminology (1); Varahamihira's Brihatsamhita (1); Arthashastra land assessment (1); Vrikshayurveda (1); Sindhu-Sarasvati conservation methods (1).	1+1+1+1+1

Mistakes to avoid:

- **Weathering** breaks rocks in place; **erosion** carries material away by water, wind or ice. They are not the same process.
- Convergent boundary = plates move towards each other (Himalaya); divergent = move apart (mid-ocean ridges); transform = slide past (San Andreas Fault).
- Oceanic crust is thinner (about 5-7 km) than continental crust (about 30-40 km); the mantle is the thickest layer and the inner core the densest.

- Sea-coast sequence is cave, arch, stack; do not mix glacial features (cirque, fjord, U-valley, moraine, arete) with river features (V-valley, meander, oxbow lake, delta).
- A delta is a deposit at the river mouth in the lower course; an oxbow lake is a cut-off meander; waterfalls and V-shaped valleys belong to the upper course.
- Barchans and ventifacts are desert (wind) features and stalactites are limestone features; the Zabo system is from Nagaland and terraces or bunds help check erosion.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND